WGSpipeline → Nextflow

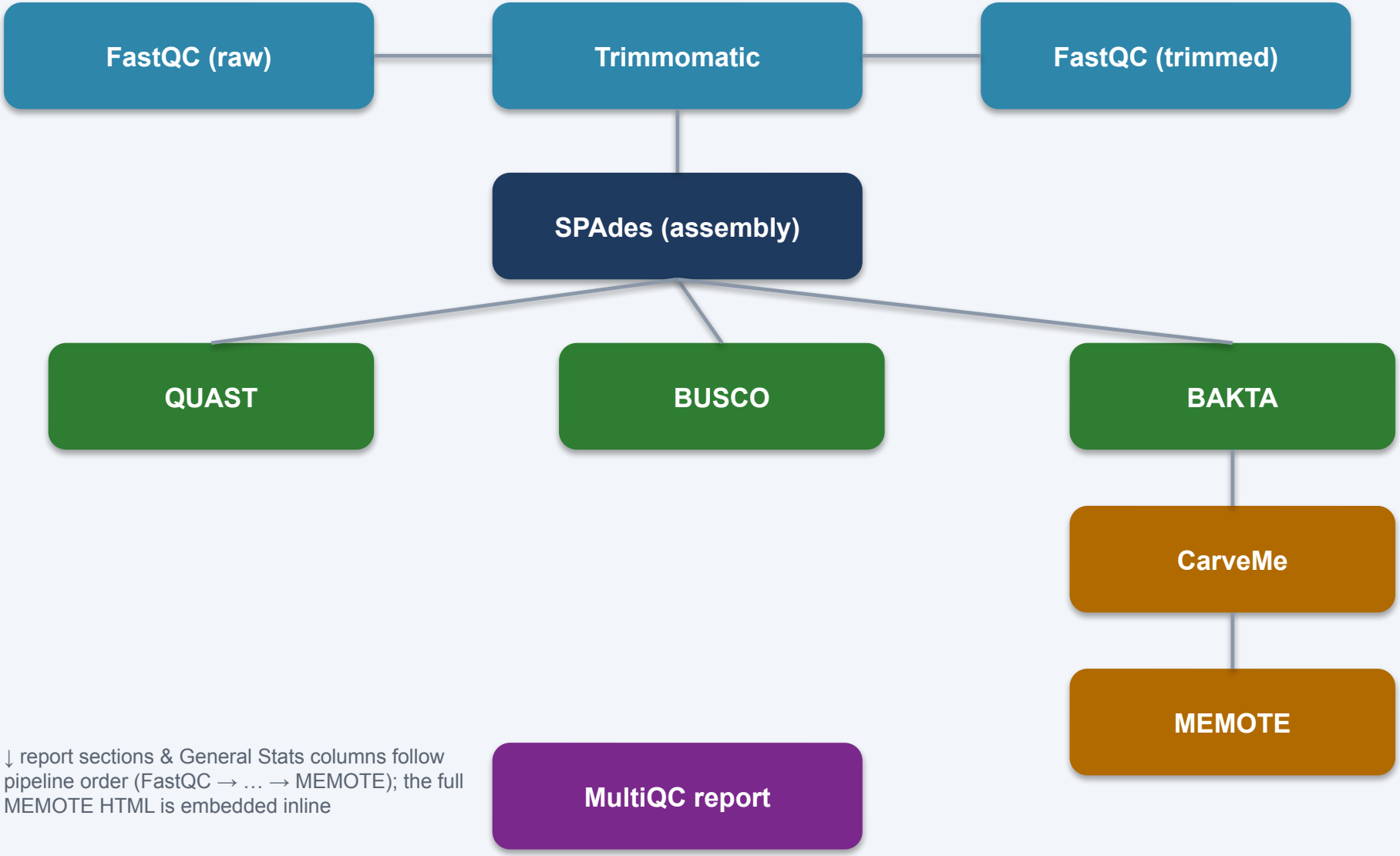
A 1:1 migration of a bacterial whole-genome-sequencing + genome-scale metabolic-modeling pipeline

Snakemake → Nextflow DSL2 · conda backend · byte-identical outputs

Background & Motivation

- Bacterial WGS + metabolic-modeling pipeline: raw reads → QC → de novo assembly → annotation → genome-scale metabolic model (GEM) → QC report.
- Original implementation: a single Snakemake file driving 10 rules, conda-managed environments, macOS/arm64 workarounds baked in.
- **Migration goal — strict parity, not a rewrite:**
- Functional 1:1 equivalence across all 10 steps and the output directory tree.
- Zero-change reuse of the 8 conda env YAMLs, helper scripts, and MultiQC config (byte-identical copies).
- Byte-identical outputs where the tool is deterministic (Trimmomatic, SPAdes, QUAST).
- All historical pitfalls preserved with the same countermeasures.
- Validation set: two E. coli isolates — K-12 (SRR1770413) and O157:H7 (SRR10194530).

Pipeline Overview — 10 Steps



↓ report sections & General Stats columns follow pipeline order (FastQC → … → MEMOTE); the full MEMOTE HTML is embedded inline

Migration Approach

- Single main.nf + nextflow.config — mirrors the original “one snakefile” convention.
- conda execution backend; all 8 environment YAMLs reused byte-identical (macOS/arm64 workarounds preserved).
- Path anchoring via $projectDir (the lesson of the original wfpath()).
- **Semantic equivalents:**
- Snakemake --keep-going → process.errorStrategy = 'finish'.
- Snakemake cache → nextflow -resume (task hash cache).
- --report_tag → multi-report filenames that coexist without overwriting.
- db/version.json sentinel → automatic re-annotation on database upgrade.
- Key assets reused byte-identical: 8 env YAMLs, multiqc_extras.py, memote_report.py, the BAKTA AMR-guard wrapper, multiqc_config.yaml.

Results — Byte-Identical vs. the Snakemake Baseline

Trimmomatic (8 .fastq.gz)	byte-identical
SPAdes contigs.fasta (2)	byte-identical
QUAST report.tsv (2)	byte-identical
BUSCO short_summary (2)	identical scores — 440/440 complete
BAKTA annotation .tsv (2)	identical line counts — 4,891 / 5,496
CarveMe diamond .tsv (2)	identical line counts — 8,961 / 9,306
MEMOTE memote_summary.tsv	identical (1 sample) / Δ0.1 (FBA tolerance)

Deterministic stages reproduce the baseline exactly — proof of a faithful migration, not an approximation.

Biological Accuracy of the Output

- BUSCO (enterobacterales_odb10): 100% complete (440/440), 0 missing, 0 fragmented — high-quality assembly.
- Assembly metrics are consistent with the known E. coli genomes:

Strain	Total length	GC %	N50
K-12 (SRR1770413)	4.84 Mb	50.33 %	159 kb
O157:H7 (SRR10194530)	5.40 Mb	50.34 %	244 kb

O157:H7 is correctly larger than K-12 (prophages / O-islands).

MEMOTE GEM scores: 68.0 (K-12) / 67.7 (O157:H7) — typical of draft genome-scale models.

Genome sizes match the sample metadata (species/strain) — no sample mix-ups.

Pitfalls Preserved — Migration Fidelity

- QUAST --labels <sample> — without it every assembly is “contigs” and samples overwrite in MultiQC.
- BAKTA AMR-guard wrapper — upstream ncbi/amr#191: one protein with overlapping NCBIFAM hits aborts annotation; guard retries chunk-wise. Triggered and verified on O157:H7.
- CarveMe staged-copy — carve derives its .tsv from the input path; feeding BAKTA’s .faa would silently overwrite BAKTA’s annotation table. Verified: BAKTA .tsv header intact.
- FastQC false-failure guard — “|| [-f output.html]” for macOS non-zero-exit-with-successful-output.
- BUSCO / MEMOTE explicit $CONDA_PREFIX/bin/python3 — shebang/PATH resolution traps.
- MultiQC cohort = the sample sheet — explicit per-sample files, never a directory scan (subset reports stay consistent).

Lessons from Nextflow 26.x (strict syntax)

- Top-level statements are forbidden when declarations exist → sample-sheet validation moved into workflow.main.
- workflow.onComplete {} removed → onComplete: section inside the entry workflow.
- publishDir string form cannot resolve input vars → closure path + saveAs to strip the out/ prefix.
- Channel.of(list) no longer splats → Channel.fromList; closure variables need .call().
- -stub runs real scripts unless a stub: block exists; dry-runs publish empty placeholders.
- Runtime params mutation in workflow.main is invisible to process directives (tag/output) → pass computed values as channel inputs.
- 20 findings (F1–F20) documented and folded back into the design doc.

Usability & Operations

nextflow run .

one command; conda environments built automatically

- nextflow run . -resume — cached restart / crash recovery.
- nextflow run . -profile lowmem — thread override (set-threads analogue).
- nextflow run . --report_tag cohortB — coexisting reports per cohort.
- CLI --key=value overrides any param; samples.tsv keeps the original format.
- Report is presentation-ready: pipeline-ordered sections/columns, full MEMOTE HTML embedded inline.
- All data outputs publish to the same directory tree as the Snakemake version.

Operational note: keep ≥ 20 GB free disk for SPAdes/BUSCO/Bakta; add a Nextflow README (current gap).

Summary

- Full 10-step bacterial WGS + GEM pipeline migrated from Snakemake to Nextflow DSL2.
- Byte-identical outputs on deterministic stages; identical metrics elsewhere.
- Biological validity confirmed (BUSCO 100%, strain-correct assembly sizes, sensible GEM scores).
- All historical pitfalls carried over with verified countermeasures.
- Production-ready entry point with -resume, profiles, and parameter overrides.

Next steps: README, optional Docker/Singularity profile, runtime test of single-sample failure isolation.